ReMixer: Object-aware Mixing Layer for Vision Transformers

Hyunwoo Kang*, **Sangwoo Mo***, **Jinwoo Shin** (* equal contribution) / Korea Advanced Institute of Science and Technology (KAIST)

TL;DR. We design a novel reweighting scheme for patch-based models considering object structures of images

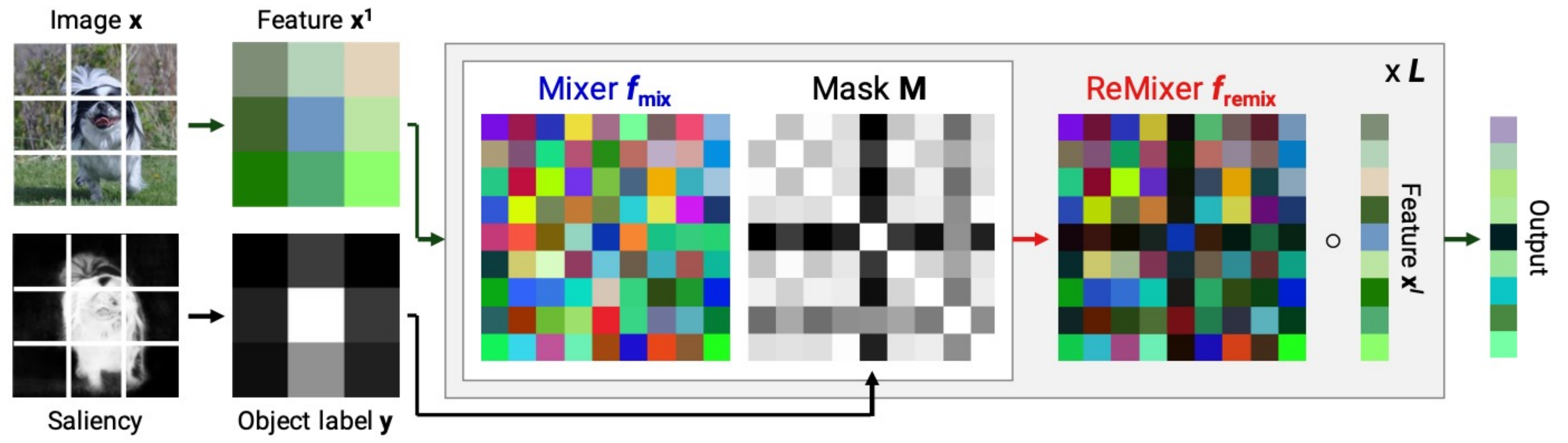
Motivation: Why we need ReMixer?

Vision Transformers (ViTs) and patch-based models perform well on various tasks, alternating the classic convolutional neural networks (CNNs)

While initial patch-based models aimed at a fully data-centric approach, recent works reveal that incorporating inductive biases (e.g., spatiality) is still essential

Unlike prior works only considering the sample-agnostic spatial inductive bias, we aim to incorporate the sample-specific object structures of images

Concept: Reweighting patch mixing layers



Patch-based models treat an image as a set of patches (or tokens) $\mathbb{R}^{N\times D}$, and apply
(a) patch mixing $f_{\text{mix}}: \mathbb{R}^N \to \mathbb{R}^N$ (i.e., mix the features among patches) and
(b) channel mixing $g_{\text{mix}}: \mathbb{R}^D \to \mathbb{R}^D$ (i.e., mix the features among channels) layers

ReMixer aims to remix the patch mixing layers based on the patch-wise object labels. Formally, let $\mathbf{M}^l \in \mathbb{R}^{N\times N}$ be a reweighting mask whose element is a reverse distance between object labels $\mathbf{y} \in \mathbb{R}^{N\times K}$ of two patches, i.e.,

$$\mathbf{M}^l_{ij} := \exp(-\kappa^{(l)} \cdot d(\mathbf{y}_i, \mathbf{y}_j)) \in (0, 1] \tag{1}$$

where $\kappa^{(l)}$ is a learnable scale parameter and $d(\cdot,\cdot)$ is a distance (e.g., ℓ_2) function

Here, the object labels are given by weakly-supervised (ReLabel) or unsupervised (BigBiGAN) manners, i.e., no additional human annotations are required

Conceptually, if $f_{\text{mix}} := \mathbf{L}_{\text{mix}}$ is linear, ReMixer reweights the layer $f_{\text{remix}} := \mathbf{M} \odot \mathbf{L}_{\text{mix}}$.
We further propose techniques to consider nonlinear patch interactions

Method: How to apply ReMixer to my models?

Using the reweighting mask $\mathbf{M}$, we calibrate the interactions of patch mixing layers.

Self-attention. Patch mixing layer is

$$f_{\texttt{mix}}(\mathbf{x}) := \underbrace{\texttt{Softmax}(\frac{\mathbf{QK}^T}{\sqrt{D_K}})}_{\text{attention matrix } \mathbf{A}} \cdot \mathbf{V} \tag{2}$$

where $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ are query, key, and value, respectively, and D_K is the dimension of $\mathbf{K}$. Here, we reweight the attention matrix, i.e.,

$$f_{\texttt{remix}}(\mathbf{x}) := [\tilde{\mathbf{A}}_{ij}] \cdot \mathbf{V} = [\frac{\mathbf{M}_{ij} \cdot \mathbf{A}_{ij}}{\sum_j \mathbf{M}_{ij} \cdot \mathbf{A}_{ij}}] \cdot \mathbf{V} \tag{3}$$

Feed-forward. Patch mixing layer is

$$f_{\texttt{mix}}(\mathbf{x}) := \mathbf{W}_m \cdot \sigma(\mathbf{W}_{m-1} \cdot \sigma(\cdots \sigma(\mathbf{W}_1 \cdot \mathbf{x}))) \tag{4}$$

where $\mathbf{W}_i$ are weight matrices and σ is a nonlinear activation. Here, we decompose it to a linear approx. $\mathbf{L}_{\text{mix}} \cdot \mathbf{x} \approx f_{\text{mix}}(\mathbf{x})$ and the residual term. Then, we reweight the linear approx. term while keeping the residual term, i.e.,

$$f_{\texttt{remix}}(\mathbf{x}) := \underbrace{(\mathbf{M} \odot \mathbf{L}_{\texttt{mix}}) \cdot \mathbf{x}}_{\text{masked linear}} + \underbrace{(f_{\texttt{mix}}(\mathbf{x}) - \mathbf{L}_{\texttt{mix}} \cdot \mathbf{x})}_{\text{residual}} \tag{5}$$

Empirically, simply dropping nonlinear activations gives a good linear approx.:

$$\mathbf{L}_{\texttt{mix}} := \mathbf{W}_m \, \mathbf{W}_{m-1} \cdots \mathbf{W}_1 \in \mathbb{R}^{N\times N} \tag{6}$$

Convolution. Patch mixing layer $f_{\texttt{mix}}(\mathbf{x}) := \mathbf{W}_{\texttt{conv}} * \mathbf{x}$ can be reformulated as a (linearized) matrix multiplicative form

$$f_{\texttt{mix}}(\mathbf{x}) = \mathbf{W}_{\texttt{linear}} \cdot \tilde{\mathbf{x}} \tag{7}$$

Then, we reweight the linearized weight matrix, i.e.,

$$f_{\texttt{remix}}(\mathbf{x}) := (\mathbf{M} \odot \mathbf{W}_{\texttt{linear}}) \cdot \tilde{\mathbf{x}} \tag{8}$$

Others. While we focus on ViT in this paper, our principle can be applied for arbitrary patch-based models, e.g., by decomposition as in the feed-forward

Results: What is the advantages of ReMixer?

Main results. Learning disentangled object representation, ReMixer…

Improves classification accuracy (e.g., ImageNet top-1 accuracy)

	MobileViT-XXS	DeiT-T	DeiT-S	DeiT-B
# params	1.3M	5M	22M	86M
Baseline	62.56	71.19	79.30	78.45
+ TokenLabeling	62.34 (-0.22)	71.61 (+0.42)	80.20 (+0.90)	81.17 (+2.72)
+ ReMixer (ours)	**64.78** (+2.22)	**74.26** (+3.07)	**81.26** (+1.96)	**82.18** (+3.73)

Improves background robustness (e.g., Background Challenge metrics)

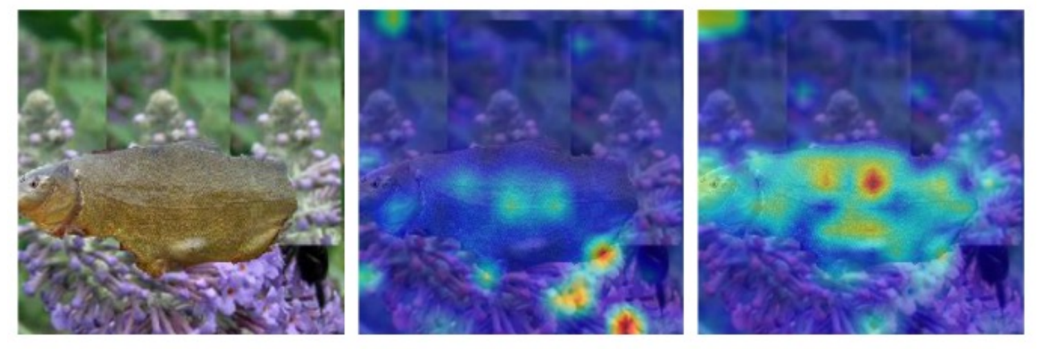
	TL	Original (↑)	Only-BG-B (↓)	Mixed-Same (↑)	Mixed-Rand (↑)	BG-Gap (↓)
DeiT-S	-	82.69	39.46	73.88	50.49	23.39
+ ConViT	-	85.51 (+2.82)	38.94 (-0.52)	76.40 (+2.52)	54.37 (+3.88)	22.03 (-1.36)
+ ReMixer (ours)	-	86.32 (+3.63)	31.78 (-7.68)	78.44 (+4.56)	56.74 (+6.25)	21.70 (-1.69)
+ ReMixer + ConViT (ours)	-	88.20 (+5.51)	30.79 (-8.67)	79.16 (+5.28)	60.52 (+10.03)	18.64 (-4.75)
DeiT-S	✓	88.52	38.00	79.43	59.36	20.07
+ ReMixer (ours)	✓	91.31 (+2.79)	29.19 (-8.81)	81.78 (+2.35)	66.92 (+7.56)	14.86 (-5.21)
MLP-Mixer-S/16	✓	88.07	38.17	79.21	58.72	20.49
+ ReMixer (ours)	✓	90.49 (+2.42)	28.72 (-9.45)	81.63 (+2.42)	65.36 (+6.64)	16.27 (-4.22)
ConvMixer-S/16	✓	87.49	36.49	80.42	58.15	22.27
+ ReMixer (ours)	✓	89.09 (+1.60)	32.67 (-3.82)	82.64 (+2.22)	63.70 (+5.55)	18.94 (-3.33)

Outperforms prior methods
- Considering spatial inductive bias (e.g., ConViT, CoAtNet)
- Utilizing patch-level supervision (e.g., TokenLabeling (TL))

⇒ In particular, prior methods are not helpful for background robustness

Analysis. ReMixer…
- Automatically learns local-to-global structures (learned mask scales)
- Gives a more object-centric view (attention visualization)

	Layer 1/4	Layer 2/4	Layer 3/4	Layer 4/4
DeiT-S	1.571	0.783	0.945	0.287
MLP-Mixer-S/16	0.744	0.000	0.001	0.061
ConvMixer-S/16	0.871	2.347	1.498	0.001


(a) Original (b) DeiT-S (c) ReMixer